STATISTICAL MODELLING

Contents

RANDOM PHENOMENA AND PROBABILITY

Random Phenomena

- Deterministic phenomenon: Phenomenon whose outcome can be predicted with a very high degree of confidence
  - Example: Age of a person (using date of birth stated in Aadhaar card)
- Stochastic phenomenon: Phenomenon which can have many possible outcomes for same experimental conditions. Outcome can be predicted with limited confidence
  - Example: Outcome of a coin toss

Characterizing random phenomena

- Sources of error in observed outcomes
  - Lack of knowledge of generating process (model error)
  - Errors in sensors used for observing outcomes (measurement error)
- Types of random phenomena
  - Discrete: Outcomes are finite
    - Coin toss : {H, T}
    - Throw of a dice : {1, 2, 3, 4, 5, 6}
  - Continuous: Infinite number of outcomes
    - Body temperature measurement in deg F

Sample space, events (discrete phenomena)

- Sample space
  - Set of all possible outcomes of a random phenomenon
    - Coin Toss : S = {H, T}
    - Two coin tosses: S = {HH, HT, TH, TT}
- Event
  - Subset of the sample space
    - Occurrence of a head in first toss of a two coin toss experiment A = {HH, HT}
    - Outcomes of a sample space are elementary events

Probability Measure

- Probability measure is a function that assigns a real value to every outcome of a random phenomena which satisfies following axioms
  - $0 \leq P(A) \leq 1$ (Probabilities are non-negative and less than 1 for any event A)
  - $P(S) = 1$ (one of the outcomes should occur)
  - For two mutually exclusive events A and B
    - $P(A \cup B) = P(A) + P(B)$
- Interpretation of probability as a frequency :
  - Conduct an experiment (coin toss) N times. If N_A is number of times outcome A occurs then $P(A) = N_A/N$

Exclusive and Independent Events

- Independent events
  - Two events are independent if occurrence of one has no influence on occurrence of other
    - Formally A and B are independent events if and only if $P(A \cap B) = P(A) \times P(B)$
    - In a two coin toss experiment, the occurrence of head in second toss can be assumed to be independent of occurrence of head or tail in first toss, then P(HH) = P(H in first toss) × P(H in second toss) = $0.5 \times 0.5 = 0.25$
- Mutually exclusive events
  - Two events are mutually exclusive if occurrence of one implies other event does not occur
    - In a two coin toss experiment, events {HH} and {HT} are mutually exclusive => P(HH and HT) = P(HH) + P(HT) = $0.25 + 0.25 = 0.5$

Some rules of probability

- Following important probability rules can be proved using Venn diagrams

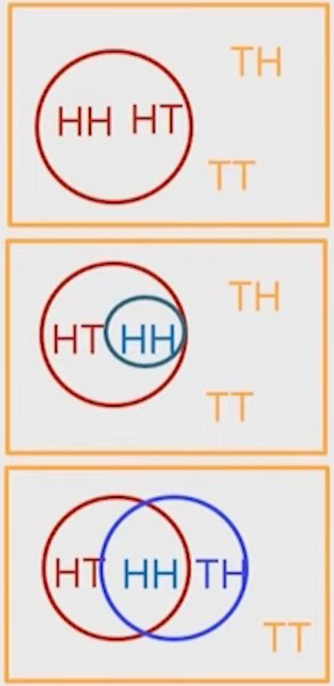
S = A = B =

All outcomes are equally likely

If A^c is the complement of event A^c,
$P(A^C) = P(S) – P(A) = 1 – P(A) = 0.5$

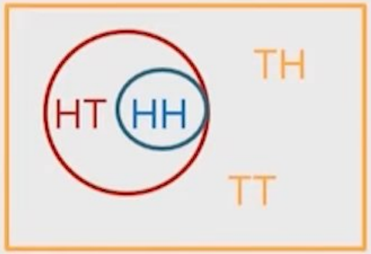
If $B \subseteq A$, $P(B) \leq P(A)$; $0.25 < 0.5$

$$P(A \cup B) = P(A) + P(B) – P(A \cap B)$$
$$= 0.5 + 0.5 – 0.5*0.5 = 0.75$$



Conditional Probability

- If two events A and B are not independent, then information available about the outcome of event A can influence the predictability of event B
- Conditional probability
  - $P(B \mid A) = P(A \cap B)/P(A)$ if $P(A) > 0$
  - $P(A \mid B)P(B) = P(B \mid A)P(A)$ - Bayes formula
  - $P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$
- Example: two (fair) coin toss experiment
  - Event A : First toss is head = {HT, HH}
  - Event B : Two successive heads ={HH}
  - Pr(B) = 0.25 (no information)
  - Given event A has occurred $Pr(B|A) = 0.5 = 0.25/0.5 = P(A \cap B)/P(A)$



Example

In a manufacturing process 1000 parts are produced of which 50 are defective. We randomly take a part from the day's production

- Outcomes : {A=Defective part B = Non-defective part}
- P(A) = 50/1000, P(B) = 950/1000

- Suppose we draw a second part without replacing the first part
  - Outcomes : {C = Defective part D = Non-defective part}
  - Pr(C) = 50/1000 (no information about outcome of first draw)
  - $P(C \mid A) = 49/999$ (given information that first draw is defective)
  - $Pr(C \mid B) = 50/999$ (given information that first draw is non-defective)
  - $P(C) = 49/999*50/1000 + 50/999*950/1000 = 50/1000$
  - $P(A \mid C) = P(A \cap C)/P(C) = P(C \mid A)P(A)/P(C) = 49/999$

RANDOM VARIABLES AND PROBABILITY MASS/DENSITY FUNCTIONS

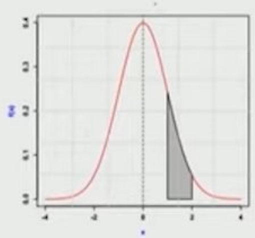
Random Variable

- A random variable (RV) is a map from sample space to a real line such that there is a unique real number corresponding to every outcome of sample space
  - eg. Coin toss sample space [H T] mapped to [0 1]. If the sample space outcomes are real valued no need for this mapping (eg. throw of a dice)
  - Allows numerical computations such as finding expected value of a RV
  - Discrete RV (throw of a dice or coin)
  - Continuous RV (sensor readings, time interval between failures)
  - Associated with the RV is also a probability measure

Probability Mass/Density Functions

- For a discrete RV the probability mass function assigns a probability to every outcome in sample space
  - Sample space of RV (x) for a coin toss experiment: [0 1].
  - $P(x = 0) = 0.5$; $P(x = 1) = 0.5$
- For a continuous RV the probability density function *f(x)* can be used to assign a probability to every interval on a real line
  - Continuous RV (x) can take any value [-∞, ∞]
  - (Area under the curve)
  - Cumulative density function F(X)

$$P(a < x < b) = \int_a^b f(x)dx$$



Shaded region: $\Pr(1 \le X \le 2)$

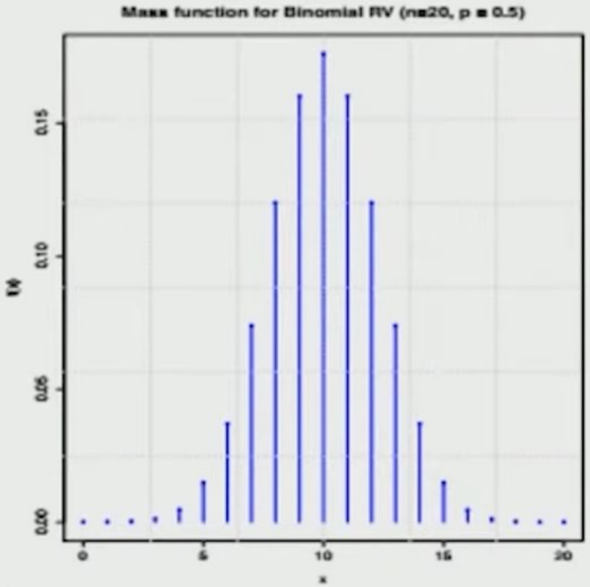
$$F(b) = P(-\infty < x < b) = \int_{-\infty}^{b} f(x)dx$$

Binomial Mass Function

- Probability of obtaining k heads in n coin tosses with p the probability of obtaining a head in any toss
- RV x represents number of heads obtained
  - Sample space : [0, 1, …n]

$$f(x=k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

  - One outcome: $\underbrace{HH\ldots H}_{k \text{ times}}\underbrace{TT\ldots T}_{n-k \text{ times}}$
  - PMF characterized by one parameter p
  - For large n it tends to a Gaussian distribution



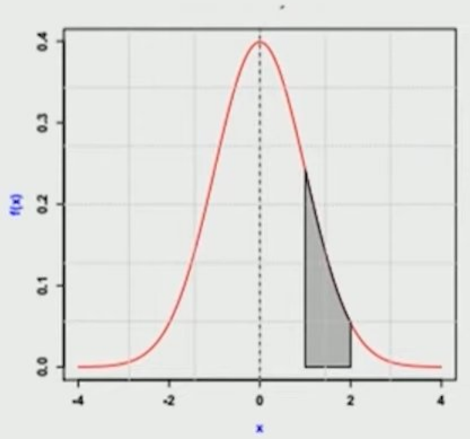
Binomial mass function for n = 20, p = 0.5

Gaussian or Normal Density Function

- Distribution used to characterize random errors in data

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  - PDF characterized by two parameters μ and σ
  - Density function is symmetric
  - Standard normal distribution $\mu = 0$ and $\sigma = 1$

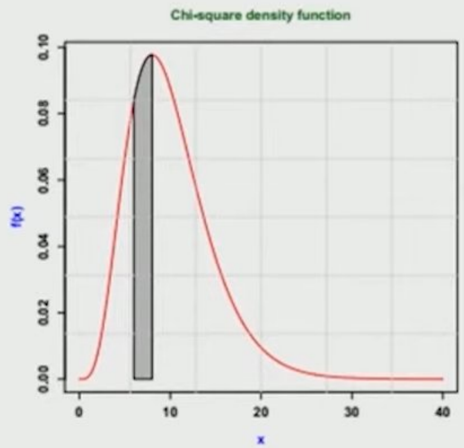


Shaded region: $\Pr(1 \le X \le 2)$

Gaussian density function for $\mu = 0$, $\sigma = 1$

Chi-square density function

- $f(x) = \frac{1}{2^{\frac{n}{2}}\Gamma(\frac{n}{2})} x^{n/2-1} e^{-x/2}$
- Density is characterized by parameter n (degrees of freedom)
- Distribution of sum of squares of n independent standard normal RVs
- Distribution of sample variance



Shaded region: $\Pr(6 \le X \le 8)$

Moments of a pdf

- Similar to describing a function using derivatives, a pdf can be described by its moments
  - For continuous distributions
    - $E[x^k] = \int_{-\infty}^{\infty} x^k f(x) dx$
  - For discrete distributions
    - $E[x^k] = \sum_{i=1}^{N} x_i^k p(x_i)$
- Mean : $\mu = E[x]$
- Variance : $\sigma^2 = E[(x-\mu)^2] = \mathrm{E}[x^2] - \mu^2$
- Standard deviation = Square root of variance = σ

Properties of Gaussian RVs

- For a Gaussian RV x
  - Mean : $E[x] = \mu$
  - Variance : $E[(x-\mu)^2] = \sigma^2$
  - Symbolically $x \sim \mathcal{N}(\mu, \sigma^2)$
- Standard Gaussian RV $z \sim \mathcal{N}(0,1)$
- If $x \sim \mathcal{N}(\mu, \sigma^2)$ and $y = ax + b$ then
  - $y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$
- Standardization
  - If $x \sim \mathcal{N}(\mu, \sigma^2)$, then $z = \frac{(x-\mu)}{\sigma} \sim \mathcal{N}(0,1)$

Computation of Probability using R

- Function to compute probability given a value X
- Lower tail probability = $P(-\infty < x < X) = \int_{-\infty}^{X} f(x)dx$
- Functions p*norm*(X, mean, std, 'lower.tail' = TRUE/FALSE)
  - *norm* refers to the distribution and can be replaced by other distributions (chisq, exp, unif)
  - X is the value (limit)
  - Parameters of the distribution (eg. mean and std for normal distribution)
  - lower.tail = TRUE (default) to obtain lower tail probability and FALSE to obtain upper tail probability

Other functions in R

- Function to compute X given probability p
  - Function q*norm*(p, mean, std, 'lower.tail' = TRUE/FALSE)
  - Lower tail probability $= P(-\infty < x < X) = \int_{-\infty}^{X} f(x)dx = p$
- Function d*norm* to compute density function value
- Function r*norm* to generate random numbers from the distribution

Joint pdf of two RVs

- Joint pdf of two RVS x and y: f(x,y)
  - $P(x \leq a, y \leq b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f(x,y)dxdy$
  - Covariance between x and y: $\sigma_{xy} = E[(x - \mu_x)(y - \mu_y)]$
  - Correlation between x and y: $\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$
- Two RVs x and y are uncorrelated if $\sigma_{xy} = 0$
- Two RVs x and y are independent if $f(x,y) = f(x)f(y)$

Multivariate Normal Distribution

- A vector of RVs $\mathbf{x} = [x_1 \quad x_2 \quad \cdots \quad x_n]^T$
- Multivariate Gaussian Distribution : $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
  - $\mathrm{E}[\mathbf{x}] = \boldsymbol{\mu}$: Mean vector
  - $\mathrm{E}[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}] = \boldsymbol{\Sigma}$: Variance-covariance matrix
  - $f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$: pdf
- Structure of Σ

$$\Sigma = \begin{bmatrix} \sigma_{x_1}^2 & \sigma_{x_1 x_2} & \cdots & \sigma_{x_1 x_n} \\ \sigma_{x_2 x_1} & \sigma_{x_2}^2 & \cdots & \\ \vdots & \vdots & \vdots & \\ \sigma_{x_n x_1} & \cdots & \cdots & \sigma_{x_n}^2 \end{bmatrix}$$

SAMPLE STATISTICS

Need for sampling

- PDFs of RVs establishes theoretical framework. But
  - Entire sample space may not be known
  - Parameters of distribution may not be known
- From a finite sample derive conclusions about the pdf and its parameters
- Sample (or observation) set is assumed to be sufficiently representative of the entire sample space
  - Proper sampling procedures and design of experiments to be used for obtaining the sample

Basic Concepts

- Population: Set of all possible outcomes of a random experiment characterized by f(x)
- Sample set (realization) : Finite set of observations obtained through an experiment
- Inference: Conclusion derived regarding the population (pdf, parameters) from the sample set
  - Inference made from a sample set is also uncertain since it depends on the sample set which is one of many possible realizations

Statistical Analysis

- Descriptive Statistics (Analysis)
  - Graphical : Organizing and presenting the data (eg. box plots, probability plots)
  - Numerical: Summarizing the sample set (eg. mean, mode, range, variance, moments)
- Inferential
  - Estimation: Estimate parameters of the pdf along with its confidence region
  - Hypotheses testing: Making judgements about f(x) and its parameters

Measures of Central Tendency - Mean

- Represent sample set by a single value
  - Mean (or average): $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$
    - Best estimate in least squares criterion
    - Unbiased estimate of population mean: $E[\bar{x}] = \mu$
    - Affected by outliers
    - Eg: Sample heights of 20 cherry trees

    [55 55 59 60 63 65 66 67 67 67 71 71 72 73 75 75 78 81 82 83]
    - Mean = 69.25 (population mean used to generate random sample was 70)
    - Mean = 71.75 (after a bias of 50 was added to first sample value)

Measures of Central Tendency – Median

- Represent sample set by a single value
  - Median: Value of x_i such that 50% of the values are less than x_i and 50% of observations are greater than x_i
    - Robust with respect to outliers in data
    - Best estimate in least absolute deviation sense
    - Eg: Sample heights of 20 cherry trees

    [55 55 59 60 63 65 66 67 67 67 71 71 72 73 75 75 78 81 82 83]
    - Median = 69 (population mean used to generate random sample was 70)
    - Median = 69 (after a bias of 50 was added to first sample value)

Measures of Central Tendency -Mode

- Represent sample set by a single value
  - Mode: Value that occurs most often (Most probable value)
  - eg. Sample heights of 20 cherry trees

[55 55 59 60 63 65 66 67 67 67 71 71 72 73 75 75 78 81 82 83]

  - Mode: 67 (three occurrences)

Measures of Spread

- Represents spread of sample set
  - Sample variance : $s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$
    - Unbiased estimate of population variance : $E[s^2] = \sigma^2$
    - Standard deviation is sqrt of variance
  - Mean absolute deviation : $\bar{d} = \frac{1}{N}\sum_{i=1}^{N}|x_i - \bar{x}|$
  - Range : $R = x_{max} - x_{min}$
  - Eg. Sample heights of 20 cherry trees

s^2 = 70.5132 and 212.25 with outlier

s = 8.392 (population std used for generating numbers was 10)

MAD = 6.85 and 9.5 with outlier

Range = 83 - 55 = 28

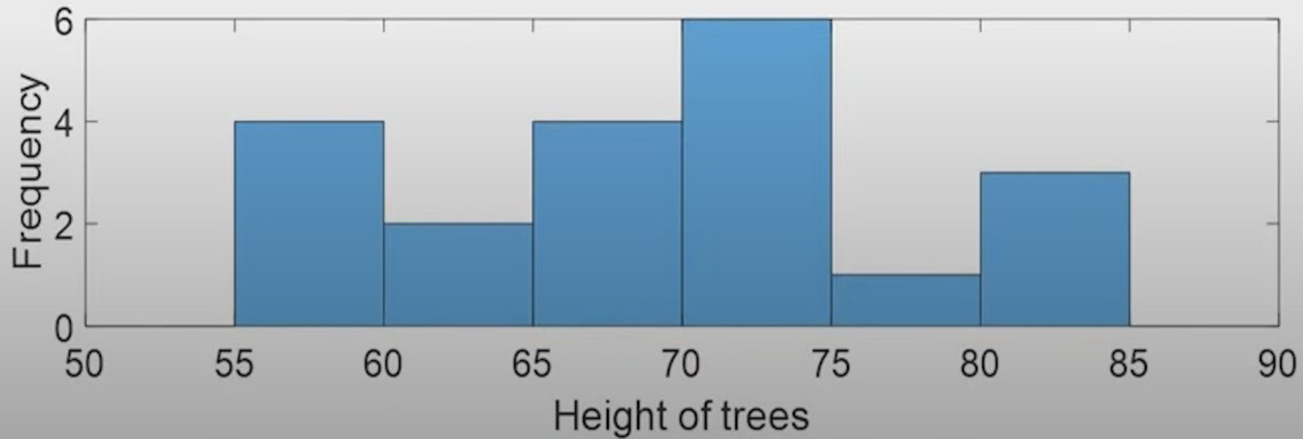
Distribution of sample mean and variance

- Sample mean
  - For any distribution sample mean is an unbiased estimate of population mean
  - If $x_i \sim \mathcal{N}(\mu, \sigma^2)$ and all observations are mutually independent, then $\bar{x} \sim \mathcal{N}(\mu, \frac{\sigma^2}{N})$
- Sample Variance
  - For any distribution sample variance is an unbiased estimate of the population variance
  - If $x_i \sim \mathcal{N}(\mu, \sigma^2)$ and all observations are mutually independent, then $\frac{(N-1)S^2}{\sigma^2} \sim \chi^2_{N-1}$

Graphical Analysis - Histograms

- Histograms
  - Divide the range of values in sample set into small intervals and count how many observations fall within each interval.
  - For each interval plot a rectangle with width = interval size and height equal to number of observations in interval
  - eg. Sample of 20 heights of black cherry trees

[73 75 55 60 66 71 81 67 83 75 82 71 63 55 72 78 67 65 67 59]



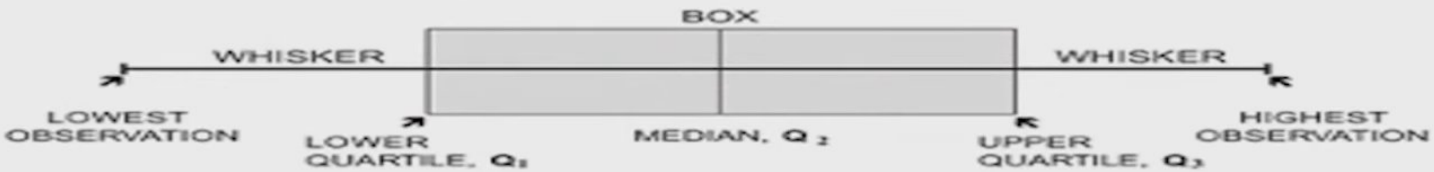
Graphical Analysis - Box Plot

- Box plot
  - Find quartiles (Q1, Q2 and Q3), minimum and maximum values in range
  - Box is between Q1 and Q3, and whiskers is between min and max values
  - eg. Sorted values of heights of 20 cherry trees

[55 55 59 60 63 65 66 67 67 67 71 71 72 73 75 75 78 81 82 83]

Q1: 64, Q2 (median): 69, Q3: 75, min: 55, max: 83

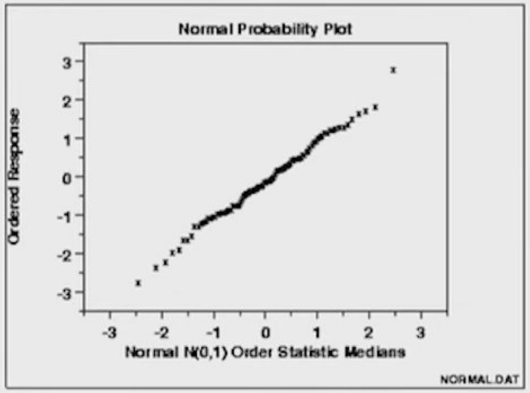
Figure 1. Box and whisker plot



Graphical Analysis – Probability Plot

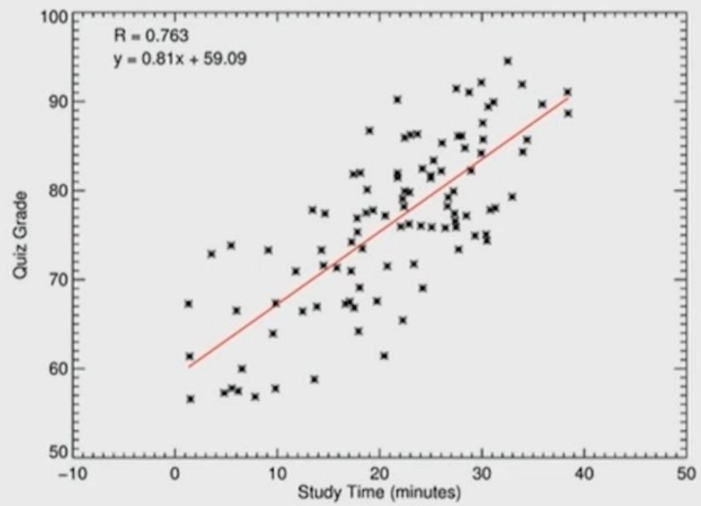
- Probability plot (p-p or q-q plot)
  - Determine different quantile values from sample set. Plot computed quantiles vs theoretical quantile values from chosen distribution
  - Same example (standardized and sorted values)

[-1.697 -1.697 -1.2206 -1.1016 -0.7443 -0.5061 -0.3870 -0.2679
-0.2679 -0.2679 0.2084 0.2084 0.3275 0.4466 0.6848 0.6848
1.0420 1.3993 1.5184 1.6374]



Graphical Analysis – Scatter Plot

- Scatter plot
  - Plot of one RV (y) against another RV (x) to examine whether there is any dependence
  - Example: Marks obtained vs study time for 100 students



Motivation for Hypotheses Testing

- Business: Will an investment in a mutual fund yield annual returns greater than desired value? (based on past performance of the fund)
- Medical: Is the incidence of diabetes greater among males than females?
- Social: Are women more likely to change mobile service provider than men?
- Engineering: Has the efficiency of the pump (η) decreased from its original value due to aging?

Hypotheses Testing

- The hypotheses is generally converted to a test of the mean or variance parameter of a population (or differences in means or variances of populations)
- A hypothesis a statement or postulate about the parameters of a distribution (or model)
  - Null hypothesis H_0 : The default or *status quo* postulate that we wish to reject if the sample set provides sufficient evidence (eg. $\eta = \eta_0$)
  - Alternative hypothesis H_1 : The alternative postulate that is accepted if the null hypothesis is rejected (eg. $\eta < \eta_0$)

Hypotheses Testing Procedure

- Identify the parameter of interest (mean, variance, proportion) which you wish to test
- Construct the null and alternative hypotheses
- Compute a test statistic which is a function of the sample set of observations
- Derive the distribution of the test statistic under the null hypothesis assumption
- Choose a test criterion (threshold) against which the test statistic is compared to reject/not reject the null hypothesis

Hypotheses Testing Procedure

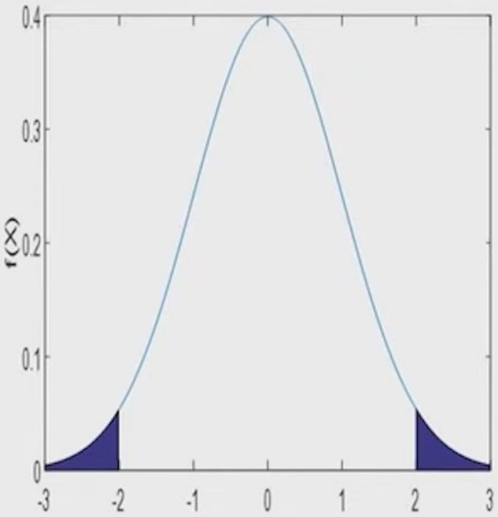
- No hypotheses test is perfect. There are inherent errors since it is based on observations which are random
- The performance of a hypotheses test depends on
  - Extent of variability in data
  - Number of observations (Sample size)
  - Test statistic (function of observations)
  - Test criterion (threshold)

Two-sided and one-sided tests

- Two sided test

  $H_0: \ \mu = 0$

  $H_1: \ \mu \neq 0$
- Test statistic standard normal RV z

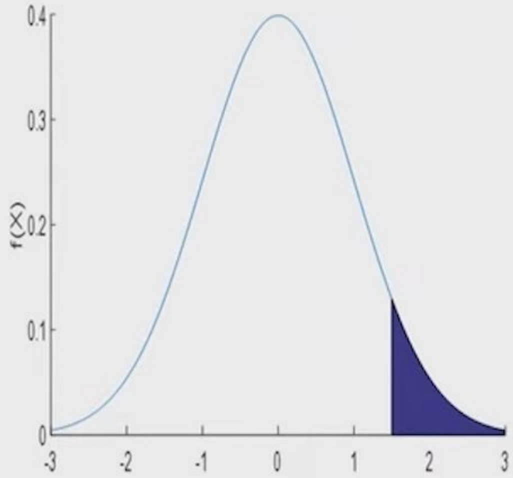


- Reject H_0 if $z \le -2$ or $z \ge 2$

- One sided test

  $H_0: \ \mu = 0$

  $H_1: \ \mu > 0$
- Test statistic standard normal RV z



- Reject H_0 if $z \ge 1.5$

Errors in Hypotheses Testing

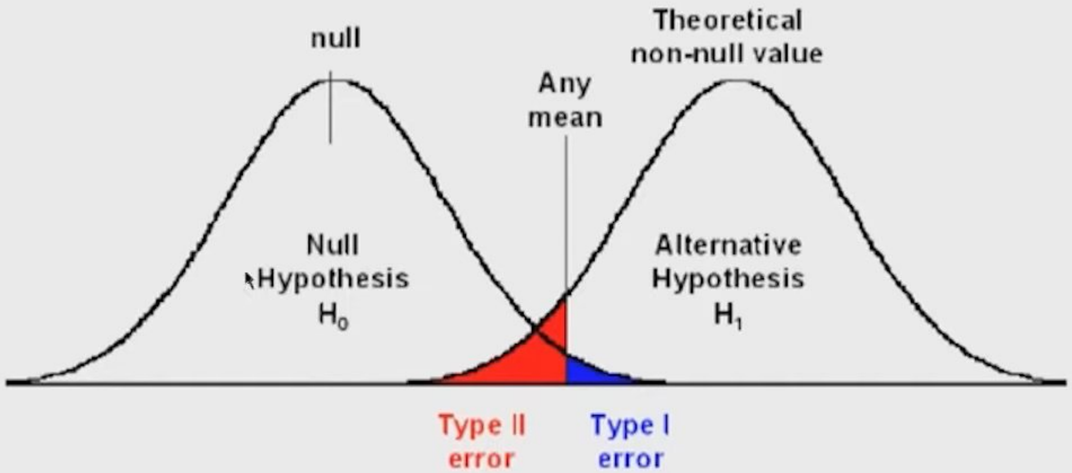
- Two Types of errors (Type I and Type II)

Decision → / Truth ↓	H_0 is not rejected	H_0 is rejected
H_0 is true	Correct Decision $Pr = 1 - \alpha$	Type I error $Pr = \alpha$
H_1 is true	Type II error $Pr = \beta$	Correct Decision $Pr = 1 - \beta$

  - Typically the Type 1 error probability α (also called as level of significance of the test) is controlled by choosing the criterion from the distribution of the test statistic under the null hypothesis

Errors in Hypotheses Testing

- Type I and Type II error probabilities



- Statistical test Power = 1 – Type II error probability
- Trade-off : If we decrease Type I error probability, then Type II error probability will increase

Test for Mean : Solid Propellant example

For a given application the burning rate of a solid propellant should be 50 cm/s.

- 25 samples of the solid propellant are taken and their burning rate noted. The average burning rate is computed to be 51.3 cm/s. The standard deviation in the burning rate is known to be 2 cm/s
- Null hypothesis : $\mu = 50$ cm/s
- Alternative hypothesis : $\mu \neq 50$ cm/s (lower or higher burning rate propellants are both unsatisfactory) – Two sided test
- Test statistic $z = \frac{\bar{x}-50}{2/\sqrt{25}} \sim \mathcal{N}(0,1)$; $z = 3.25$
- Critical value for $\alpha = 0.05$ is ± 1.96
- Decision: Reject null hypothesis

Test for Differences in Means : Training example

Two groups of teachers of similar capabilities are trained by two methods A and B. Is Method B more effective than Method A?

- 10 teachers in each group. Average scores and standard deviation of scores after training are Group 1: $\bar{x}_1 = 70, s_1 = 3.3665$ Group 2: $\bar{x}_1 = 74, s_1 = 5.3955$
- Null hypothesis : $\mu_1 - \mu_2 = 0$
- Alternative hypothesis : $\mu_1 - \mu_2 < 0$ – one sided test
- Test statistic (assuming unknown but equal variances for two groups)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_p^2}{N_1} + \frac{s_p^2}{N_2}}} \sim t_{N_1+N_2-2}; \quad S_p = \frac{(N_1-1)s_1^2 + (N_2-1)s_2^2}{N_1+N_2-2} \quad t = -1.989$$

- Critical value for $\alpha = 0.05$ is -1.73
- Decision: Reject null hypothesis (Method B is better)

Test for Differences in Variances : Process Yields

The variability in yields from two different processes are to be compared to decide whether they are identical or not

- 50 samples for each process taken. Yield variances are found to be $s_1^2 = 2.05$ and $s_2^2 = 7.64$
- Null hypothesis : $\frac{\sigma_1^2}{\sigma_2^2} = 1$
- Alternative hypothesis : $\frac{\sigma_1^2}{\sigma_2^2} \neq 1$ – two sided test
- Test statistic (assuming unknown but equal variances for two groups)

$$f = \frac{s_1^2}{s_2^2} \sim F(N_1 - 1, N_2 - 1);\ f = 0.27$$

- Critical value for $\alpha = 0.025$ is 0.567 and $\alpha = 0.975$ is 1.762
- Decision: Reject null hypothesis (Process 2 has higher variability)

Summary of useful hypotheses tests

Type of test	Characteristic	Example	Application
z-test	Sum of independent normal variables	Test for a mean or comparison between two group means (variance known)	Test coefficients of a regression model
t-test	Ratio of a standard normal variable and chi-square variables with p degrees of freedom	Test for a mean or comparison between two group means (variance unknown)	Test coefficients of a regression model
chi-square test (p degrees of freedom)	Sum of p independent standard normal variables	Test for variance	Test quality of regression model
F-test (p_1 and p_2 degrees of freedom)	Ratio of two chi-square variables	Test for comparing variances of two groups	Choose between regression models having different number of parameters